

Title of Project

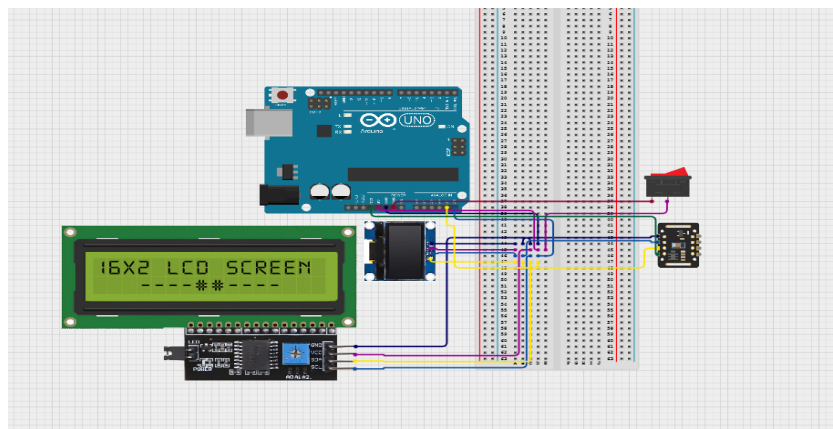
Principle / Concept

- The MAX30102 works on **photoplethysmography (PPG)**, detecting blood volume changes using infrared and red LEDs.
- Arduino processes these optical signals and converts them into heart rate and oxygen saturation values.

Advantages / Project Outcomes

- Portable and low-cost device suitable for continuous real-time monitoring.
- Provides accurate, immediate health data without requiring large medical equipment.

Block Diagram /Circuit Diagram

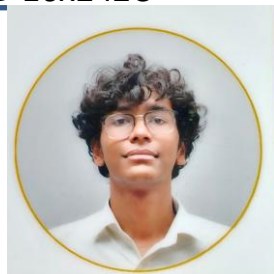


Actual Photograph of Project / Prototype



Component Used

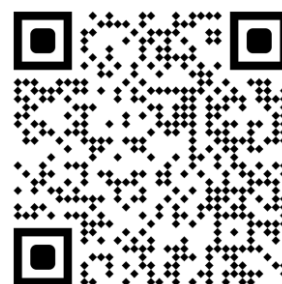
- Arduino UNO
- MAX30102 Pulse Oximeter Sensor
- OLED Display (I2C)
- Jumper wires / USB cable
- LCD 16x2 I2C



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Applications

- Useful in hospitals, clinics, rural health centers, and emergency health screening.
- Ideal for personal daily health tracking and fitness monitoring.